

Manual Execution Guide (PNPM)

A complete reference on configuring, migrating, testing, and running the full Chokro monorepo using **pnpm** instead of Docker.

1. Quick Start (All Setup & Run Commands Together)

To set up the database and boot both the backend API and mobile app concurrently from scratch:

```
# 1. Duplicate environment template
cp .env.example .env

# 2. Install dependencies across all monorepo workspaces
pnpm install

# 3. Push schema to PostgreSQL, run migrations, and seed initial dataset
pnpm db:setup

# 4. Start all applications concurrently (Next.js API/Admin + Expo bundler)
pnpm dev
```

TURBO CONCURRENT DEV: pnpm dev starts **@chokro/api** on <http://localhost:3000> and **@chokro/mobile** on Expo Metro bundler concurrently.

2. Prerequisites & Environment Configuration

Requires **Node.js** >= 22.13.0 and **pnpm** >= 11.18.0. Update `.env` with your local PostgreSQL credentials:

```
DATABASE_URL=postgres://postgres:postgres@localhost:5432/chokro
JWT_SECRET=replace-with-a-long-random-jwt-secret
QR_SECRET=replace-with-a-long-random-qr-secret
EXPO_PUBLIC_API_URL=http://localhost:3000
```

3. Database Lifecycle & Migrations (packages/db)

Database operations use **Drizzle ORM** and PostgreSQL:

```
pnpm db:setup # Complete lifecycle (push schema + run migrations + seed data)
pnpm db:push # Push schema changes directly to DB (drizzle-kit push)
pnpm db:migrate # Run migration runner (packages/db/src/migrate.ts)
pnpm db:seed # Seed test accounts, rate cards, and drop zones
```

4. Running Services Individually (Multi-Terminal)

For debugging specific parts of the system in separate terminal sessions:

Backend REST API & Web Admin Console (apps/api)

```
# Starts Next.js 16 server on port 3000 (Admin console at http://localhost:3000/admin)
pnpm --filter @chokro/api dev
```

Mobile Client (apps/mobile - Expo SDK 57 / React Native 0.86)

```
pnpm --filter @chokro/mobile start # Expo Metro bundler & interactive terminal
pnpm --filter @chokro/mobile web # Web browser preview
pnpm --filter @chokro/mobile ios # iOS Simulator (macOS required)
pnpm --filter @chokro/mobile android # Android Emulator
```

5. Production Build & Execution

```
pnpm build # Build all workspaces with Nx / Turborepo
pnpm --filter @chokro/api start # Start Next.js production standalone server
```

6. Tests, Typechecking & Code Quality

API integration tests use an embedded PGLite (WASM Postgres) engine. **No running Docker or Postgres daemon is required** to run tests!

```
pnpm test # Run full Jest test suite (zero DB setup needed)
pnpm typecheck # Typecheck entire monorepo with TypeScript
pnpm lint # Run ESLint validation across all apps and packages
```

7. Command Quick Reference Summary

Target / Workflow	Command	Description
Full Monorepo Dev	<code>pnpm dev</code>	Runs API backend (port 3000) + Expo mobile bundler in parallel
API & Web Admin	<code>pnpm --filter @chokro/api dev</code>	Next.js 16 REST API + Admin Console at http://localhost:3000/admin
Mobile App (Expo)	<code>pnpm --filter @chokro/mobile start</code>	Starts Expo Metro bundler with QR code for Expo Go / simulator
Database Setup	<code>pnpm db:setup</code>	Pushes schema, runs migrations, and seeds baseline data in one shot
Automated Tests	<code>pnpm test</code>	Runs Jest suites against in-memory PGLite DB (zero config)
Typecheck	<code>pnpm typecheck</code>	Validates TypeScript types across all workspaces
Production Build	<code>pnpm build</code>	Builds production artifacts using Nx build pipeline